Advanced Programming Practice · 21CSC203P

Unit 1 · Session 3 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 3 : SLO – 1 | Bohm-Jacopini Theorem

Activity (Solutions)

• 1) Three control structures in Java — SEQUENCE: int a = 5; int b = 10; int c = a + b; (statements run in order). SELECTION: if (c > 10) System.out.println("big"); else System.out.println("small"); ITERATION: for (int i = 0; i < 3; i++) System.out.println(i); — Bohm-Jacopini theorem: any algorithm can be expressed using only these three structures.

• 2) goto-like (nested if-break) converted to structured code — Unstructured: outer: while(true){ read(x); if(x<0) break outer; if(x==0) continue; process(x); } Structured: int x = read(); while (x >= 0) { if (x != 0) process(x); x = read(); } — same logic using a loop condition instead of break-to-label.

• 3) Importance of structure: single-entry/single-exit blocks make control flow predictable, so code is easier to read, debug, test and maintain; it eliminates spaghetti code caused by arbitrary jumps, enables modular reasoning, and lets compilers optimize better.